

ECE 551

HW4 *(100 pts)*

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- Due Weds March 23rd @ 11:55PM
 - Work Individually
 - Use descriptive signal names
 - Comment & indent your code
 - Code will be judged on coding style

HW4 Problems 1&2 (10pts) + (10pts)

1. **(10pts)** Complete the Synopsys Design Vision tutorial. Sign below, (preferably in blood).

I, _____ completed the Synopsys Design Vision tutorial. If I had any problems with it, I discussed them with the TA or Instructor, either in person, or through email.

2. **(10pts)** Project Team Formation

Form a 4 person project team, **Come up with a team name**, and fill in the table below:

Team Name	
Person1	
Person2	
Person3	
Person4	

HW4 Problem 3 (25pts) UART_wrapper/RemoteComm

Started as Exercise14 on Mon March 9th

This problem has to do with the completion of **UART_wrapper** and **RemoteComm** used for receiving navigation commands.

See Exercise 14 for complete specification.

Submit:

- **UART_wrapper.sv**
- **RemoteComm.sv**
- **CommTB.sv**
- Proof your self-checking **commTB** passed. Make sure your username can be clearly seen in the images.

HW4 Problem 4 (35pts) SPI Transceiver

Started as Exercise15 on Mon March 16th

This problem has to do with the completion of **SPI Transceiver** used for interfacing with peripheral chips such as Gyro and A2D converter.

See Exercise 15 for complete specification.

Submit:

- **SPI_main.sv**
- **SPI_main_tb.sv**
- Proof your self-checking passed. Make sure your username can be clearly seen in the images.

HW4 Problem 5 (20pts) Synthesize your UART

Started as Exercise16 on Fri March 20th

- In Ex14 your **UART_tx** and **UART_rx** were combined to produce a UART transceiver (**UART.sv**). You will now synthesize **UART.sv** and its children (**UART_tx** & **UART_rx**).
- Write a synthesis script (**UART.dc**) to synthesize your **UART**. The script should perform the following:
 - Defines a clock of 500MHz frequency and sources it to clock
 - Performs a set don't touch on the clock network
 - Defines input delays of 0.4 ns on all inputs other than clock
 - Defines a drive strength equivalent to a 2-input nand of size 2 from the Synopsys 32nm library (NAND2X2_LVT) for all inputs except clock and rst_n
 - Defines an output delay of 0.4ns on all outputs.
 - Defines a 0.10pf load on all outputs.
 - Sets a max transition time of 0.15ns on all nodes.
 - Employs the Synopsys 32nm wire load model for a block of size 16000 sq microns
 - Compiles, then flattens the design so it has no hierarchy, and compiles again.
 - Produces a min_delay & max delay report
 - Produces an area report
 - Writes out the gate level verilog netlist (**UART.vg**)
- Submit:
 - Your synthesis scripts (**UART.dc**)
 - The output reports for area (**UART_area.txt**)
 - The gate level verilog netlist (**UART.vg**)